

Lecture 9

I Fund-Thm. of Arithmetic

II Loop Ends

III Review

Thm: If $n \geq 2$ is an integer, there is a unique factorization

$$n = p_1 p_2 \dots p_k \quad \text{where} \quad p_i \text{ is prime } \forall i=1 \dots k$$

and

$$p_1 \leq p_2 \leq \dots \leq p_k.$$

($\forall n$ if $n = p_1 \dots p_a = q_1 \dots q_b$ then $p_1 \dots p_a = q_1 \dots q_b$)

$$\hookrightarrow 6 = 2 \cdot 3 \\ = 3 \cdot 2$$

Proof: (Uniqueness) Assume for contradiction that uniqueness fails.

Let $S = \{ n \geq 2 \text{ integer} : \exists \begin{matrix} p_1 \leq \dots \leq p_a \\ q_1 \leq \dots \leq q_b \end{matrix} \text{ distinct prime factorizations of } n \}$.

$$S \neq \emptyset.$$

By the WOP, we can choose a least $n \in S$ such that $\leq \mathbb{Z}_+$.

$$n = \boxed{p_1 p_2 \dots p_a = q_1 q_2 \dots q_b} \quad \text{for some } a, b \in \mathbb{Z}_+ \text{ primes } p_i, q_j$$

Observe that $p_1 \mid \overbrace{q_1}^b \overbrace{q_2 \dots q_b}^c$.

If $p_1 \neq q_1$ then $\gcd(p_1, q_1) = 1$

So by Euclid's lemma, $p_1 \mid q_2 q_3 \dots q_b$.

If $p_1 \neq q_2$ then $\gcd(p_1, q_2) = 1$ and
by Euclid $p_1 \mid q_3 \dots q_b$.

Such that
 $p_1 \leq \dots \leq p_a \neq$
 $q_1 \leq \dots \leq q_b$

Repeating this argument b times, we conclude that $p_1 \mid q_i$ for some i , so $\boxed{p_1 = q_i}$ for some $i \in \{1, \dots, b\}$. $\exists i (p_1 = q_i)$

Observe that, $n' = \underbrace{p_1 p_2 \dots p_a}_{p_1} = \frac{q_1 \dots q_b}{q_i}$, so $n' < n$

has two distinct prime factorizations, violating the minimality of n .
if same, adding $\leftarrow p_1$ and q_i would make $\square$

$p_1 \leq \dots \leq p_a$ and $q_1 \leq \dots \leq q_b$ the same.

< Existence after Minkowski >.

Claim: If p is prime, $\gcd(p, (p-1)!) = 1$.

Proof: $(p-1)! = 1 \cdot 2 \cdot 3 \cdot \dots \cdot (p-1)$

has a prime factorization $q_1 q_2 \dots q_m$
where $q_i < p$ for all $i = 1 \dots m$.

By uniqueness in the FTA, all factors of $(p-1)!$
are strictly less than p . Thus, $\gcd(p, (p-1)!) = 1$.

□

II Loose Ends.

(i) $a \equiv b \pmod{m}$ & $b \equiv c \pmod{m}$
 $\rightarrow a \equiv c \pmod{m}$.

Prop 3 (ii) If $a \equiv b \pmod{m}$ and $c \equiv d \pmod{m}$ then
 $ac \equiv bd \pmod{m}$.

Proof: Assume $a, b, c, d \in \mathbb{Z}$ and $a \equiv b \pmod{m}$
 $c \equiv d \pmod{m}$.

Since $m \mid b-a$, observe that $m \mid c(b-a)$

i.e. $ca \equiv cb \pmod{m}$.

Since $m \mid c-d$, $m \mid b(c-d)$ so
 $bc \equiv bd \pmod{m}$.

Combining these, $ac \equiv bd \pmod{m}$ as desired $\square$.

Corollary: $\forall k \in \mathbb{Z}_+$ $a \equiv b \pmod{m} \rightarrow a^k \equiv b^k \pmod{m}$

(apply above k times).

2) Base b expansion. $b=2$ (binary).

ex: $47 = 7 \cdot 10^0 + 4 \cdot 10^1.$

$$47 = 2 \cdot 23 + 1$$

$$23 = 2 \cdot 11 + 1$$

$$11 = 2 \cdot 5 + 1$$

$$5 = 2 \cdot 2 + 1$$

$$2 = 2 \cdot 1 + 0$$

$$1 = 2 \cdot 0 + 1$$

Repeatedly
divide
by b

$$\begin{aligned} &1 \cdot 2^5 + 0 \cdot 2^4 \\ &+ 1 \cdot 2^3 + 1 \cdot 2^2 + 1 \cdot 2^1 \\ &+ 1 \cdot 2^0 \\ &\quad \quad \quad // \end{aligned}$$

$$47 = 2(2(2(2(2(1)+0)+1)+1)+1) + 1$$

II Review

Lec 1-3: How to read/write/understand math statements involving $\neg, \vee, \wedge, \rightarrow, P(x), \exists, \forall$

★ How to prove $\forall, \exists$ 
use $\forall, \exists$
negate $\forall, \exists$

★ "the table" metaphorical place for named bound variables + propositions which are known.

$\forall x(P(x) \rightarrow Q(x))$ $\forall x(\neg Q \rightarrow \neg P)$ $\neg \forall x(P(x))$

★ Direct Proof, Contrapositive, Contradiction, Counterexample.
Choose the strategy which puts the most on the table.

★ Structure: Beginning, Middle, End.

* A good proof is clear and correct.

Lec 4-6: Sets, function, 1-1, onto, fog, f^{-1} , $f^{-1}(S)$, Cardinality

* Difference between 1-1 and onto.

* Awareness of domains and codomains $f: \underset{A}{\mathbb{R}} \rightarrow \underset{B}{\mathbb{R}}$

* Awareness of types. "A is onto." set meaningless.

* Descriptions of enumerations/bijections to show A is countable

A is countably infinite $\iff |A| = |\mathbb{Z}_+|$

$\iff \exists f: \mathbb{Z}_+ \rightarrow A$ bijection

$\iff \exists f: A \rightarrow \mathbb{Z}_+$ bijection

$\iff \exists$ enumeration
 $a_1, a_2, a_3, \dots$ A B

ideas: $A \times B$ snake path, $A \subseteq B$ crossing out repeats, interleaving, ...

Lec 6-8: Number theory. (look at mind map).

- ★ Built everything from scratch.
- ★ A proof is a finite piece of writing which can establish the truth of infinitely many statements.

Prove or disprove: $A = \{x \in \mathbb{Z} : \underbrace{5x \equiv 3 \pmod{7}}_{P(x)}\} \subseteq \mathbb{Z}$
is countable infinite.
 $\hookrightarrow$ true is an enumeration of A .

Form a belief:

Can't be uncountable $\subseteq \mathbb{Z}$.

Are there any x : $5x \equiv 3 \pmod{7}$?

"Divide by 5" : inverse $\xrightarrow{\gcd(5,7)=1}$ Bezout.

$$\begin{array}{lcl} 7 = 1 \cdot 5 + 2 & \nearrow & 1 = 5 - 2(7-5) \\ 5 = 2 \cdot 2 + 1 & & = 3 \cdot 5 - 2 \cdot 7. \end{array}$$

$$5^{-1} \equiv 3 \pmod{7}.$$

$$\therefore x \equiv 3 \cdot 3 \equiv 2 \pmod{7}.$$

By defn: $x \equiv 2 \pmod{7} \iff x \bmod 7 = 2$

$$x = 2, 9, 16, \dots$$

$$\text{Any } x = 2 + 7k, k \in \mathbb{Z} \text{ works.}$$

Proof: Observe that $\gcd(5, 7) = 1$ so 5 has an inverse modulo 7.

Computing the Bezout coeffs, we see that $5^4 = 3 \pmod{7}$,

$$\text{so } A = \{x \in \mathbb{Z} : x \equiv 2 \pmod{7}\}.$$

A is infinite since $2 + 7k \in A$ for all $k \in \mathbb{Z}$.

Let S be an enumeration of $\mathbb{Z}$. $\dots -1, 0, 1, 2, 3, \dots$

Let S' be the sequence obtained $\dots, -5, 2, 9, 16, \dots$
by removing all $x \in \mathbb{Z}$ such that $x \not\equiv 2 \pmod{7}$.

By construction, S' is an enumeration of A . $\therefore$
 A is countably infinite. $\square$

